

Chapter 5

Metrics, Incentives, Rewards, and Culture for Impact



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5.1 Introduction

IBM's University-Industry (UI) relationships journey provides lessons for practitioners about metrics, incentives, rewards, and culture. For over 100 years, IBM's innovations in hardware, software, service & systems have transformed work processes in business, government, science, and personal productivity. Exploring science, technology, and commercial possibilities requires highly-skilled university-trained talent, and therefore IBM's Global University Program (GUP) and UI relationships have played a significant role in these achievements (Aspray & Williams, 1994; Spohrer, 2010). This chapter explores how internal and external business drivers have fostered the design and redesign of IBM's UI programs and relationships. Specifically, we reflect on what has changed and what remains the same regarding the metrics, incentives, rewards and culture to drive impact, and explore why changes are made. We also show how a framework developed at IBM can be applied at other companies.

5.2 Opportunities and Challenges

The digital transformation of business and society is accelerating, creating a wide range of opportunities and challenges for UI relationships and collaborations, some old and some new—but it is sometimes difficult to see the forest for all the trees

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(Wurth et al., 2024). Successful UI relationships have produced significant benefits that could not have been achieved by industry or universities alone. For example, in the 1940s, IBM collaborated with Harvard University on the Mark I computer and with Columbia University on the first scientific computing course, the forerunner of computer science departments around the world. Both the 1997 IBM Deep Blue chess playing system and IBM Watson Jeopardy! general knowledge answering system involved collaborations with Carnegie-Mellon University (CMU), as well as other universities. These and many other milestones in computer science history would not have been possible without cultivating highly impactful UI relationships.

A hallmark of successful UI relationships is clearly defined goals that lead to hiring talent, publicity, and commercial offerings for industry, as well as jobs for graduates, publicity, and publications for university partners. Successful practices breed more successful practices, creating a culture of productive collaborations with distinct, but aligned, incentives.

While the opportunities are significant, challenges exist as well. For example, challenges can arise on topics related to Intellectual Property (IP) (e.g., acceptance of open innovation with publications or inventions and proprietary datasets that one or both institutions has rights to commercialize). Occasionally talent disputes can also cause a “brain-drain” (e.g., if industry becomes too aggressive hiring away not just students, but top faculty in large numbers). More recently, innovative win-win solutions to address brain drain are being tested in UI collaboration models as we will see in some of the examples that will be described later in this chapter.

Successful UI relationships must be continuously adapted and re-invented as priorities and champions change on both sides. Maintaining a healthy culture of collaboration requires careful attention to ongoing practices, learning from mistakes, and making needed adjustments. For IBM, every new generation of computational capabilities, and data governance responsibilities created challenges and opportunities for UI relationships. In this chapter, we draw from these experiences to illustrate the importance of metrics, incentives, rewards, and culture.

5.3 Prior Research

Before examining the metrics, incentives, rewards, and culture for impact from an IBM experiences perspective, it is worth considering an even broader perspective from systematic literature reviews on these four key UI relationship topics. A wide range of UI organizational forms have been documented and studied as a vehicle to enhance innovation through knowledge sharing and collaboration (Ankrah & Omar, 2015). The rise of the entrepreneurial university has increased the range of collaborative (and competitive) organizational and ecosystem forms as well (Compagnucci & Spigarelli, 2020).

University startups enrich the talent ecosystem (Klofsten et al., 2019), and attract industry attention. However, many startups fail commercially because they are

based on relatively embryonic technologies, often far ahead of sustainable market demand. Nevertheless, the enriched talent ecosystem can have benefits for industry partners looking to innovate their established offerings. For example, for IT (Information Technology) companies like IBM, a key metric is university adoption of IBM hardware and software platforms. Adoption reduces the cost of recruiting and training talent as well as reduces the cost of acquiring and integrating university startups. Furthermore, the rise of the entrepreneurial university is driving changes in metrics related to UI collaborations such as technology licensing (Etzkowitz, 2016).

Quantitative and qualitative metrics across input, process, output, and impact are recommended in Perkmann et al. (2011) for UI alliance performance measurement systems. Other studies identify institutional, relationship, output, and framework factors as most relevant for success (Rybníček & Königsgruber, 2019). While historically the number of UI co-authored publications has been a strong indicator of success, metrics are constantly changing as methodological refinements are discovered and in general more and richer types of data become available as a result of the digital transformation of all industries (Tijssen et al., 2016). For example, one might ask how AI-generated text may impact the number of university-industry co-authored publications? Gaining new knowledge to solve problems remains a primary driver of UI research collaborations. However, in a competitive world, UI collaborations can also catalyze public research funding, which can be another metric (Piva & Rossi-Lamastra, 2013). For example, smart city projects are a potential source of public funding (Guenduez et al., 2024). However, as illustrated below, national competitiveness is another source of public-private funding.

Even as the number of studies and systematic literature reviews of UI metrics grows, the challenges of metrics remain numerous; the biggest challenge being the diversity of stakeholders to align, as described succinctly in Cohen et al. (2024):

It involves stakeholders with different outlooks and impact expectations. Moreover, the multidimensional nature of the impacts themselves means they are tangible and intangible, short- and long-term, direct and indirect, positive and negative, making their measurement process very complex.

Some studies have explored the motivations of academics, which is often to extend research more than commercialization (D'Este & Perkmann, 2011). Beyond individual motivations, organizational incentives also matter (Link & Siegel, 2005). Both organizational and individual incentives are important factors (Rajalo & Vadi, 2017). Pragmatic incentives have been evaluated at the national level, shedding some light on the pros and cons of UI co-authored publications metrics and incentives (Park & Leydesdorff, 2010). Only recently (with increasing amounts of longitudinal data over several industry cycles of boom and bust) have scholars actively begun to unravel some of the intricacies of the longer-term coevolution of metrics, incentives, rewards, and cultures as an ongoing dynamic in a competitive world (O'Dwyer et al., 2023).

Not surprisingly, mutual respect and trust are key ingredients for success, along with contractual safeguards (Hemmert et al., 2014), especially in emerging markets (Sassi & Mshenga, 2024). As the market power of some industry players rivals that

of nations, perceived fairness is an increasingly important issue in trust building and culture (Clauss et al., 2024).

5.4 Options, Strategies, and Impact

While the standard program options for UI relations are well established (see Table 5.1), continuously adapting UI investments to maximize the potential for impact depends on multiple factors, including business conditions and strategic focus areas. For example, during the 2000s, IBM’s business focus was on the globally integrated enterprise and growth of IBM’s service business in order to support large enterprise customers (e.g., businesses and governments, and yes, even top regional universities) on their digital transformation journeys (Palmisano, 2006; Spohrer, 2017; Lella et al., 2012). However, in the 2010s, IBM’s business focus shifted to cognitive computing and hybrid cloud (Kelly & Hamm, 2013; Krishna, 2017).

The standard program options can be viewed through different lenses. For example, depending on business conditions (resources available) as well as business focus areas (new or mature technical areas), UI strategies can be mapped to investment levels in the six Rs program areas (Spohrer, 2010):

- 1. **Research:** Creating new knowledge that can both be published and protected as IP for commercialization is typically a top strategic priority.
- 2. **Readiness:** Skills development is essential in business focus areas and also typically a top strategic priority.

Table 5.1 Program options for university-industry (UI) relations

Large resources	Embedded industry researchers Equipment access Intellectual property (IP) licensing	Grand challenge research Equipment donations Contracted research	Global university team Endowed research center Public-private partnership
Medium resources	Internships Co-authored publications Conference sponsorships	PhD fellowships Faculty awards Co-created (online) courses	Endowed faculty chair Co-created degree programs Joint grant proposal development
Small resources	Hackathons (competitions) Mentor student teams Mock interviews	Open source project Guest lecture series Curriculum advisor board	Professional association sponsor Conf./journal best paper awards Partnership executive program
Options: Resources-time	Small time (0–2 years)	Medium time (2–4 years)	Large time (>4 years)

3. Recruiting: From permanent hires to internships, these programs include a human resources element.
4. Revenue: Partnership executive programs are essential for sustaining long-term win-win relationships.
5. Responsibility: Employees donating time to mentor students and give guest lectures.
6. Regions: Programs related to public-private partnerships for regional economic development (e.g., IBM Smarter Cities).

During the 2000s and 2010s, the six Rs provided a way for IBM's GUP team of over a hundred employees in 45 nations around the world to communicate effectively with university partners, from faculty to deans to presidents and provosts (Spohrer, 2013). Inside IBM, a central database kept track of the history of investments made as well as outcomes for some 2000 institutes of higher education worldwide. With over 300,000 employees worldwide and with new hires and interns totalling around 30,000 individuals each year, it was essential to keep track of a great deal of UI information to support UI relationships and win-win strategies.

For example, an IBM university PEP (Partnership Executive Program) executive could request a report for the corporate GUP team before visiting a university to get a sense of the relationship history, including funded research faculty, number of IBM employees with degrees (and what discipline areas) from that university hired over the years, number of current active IBM employees and executive with degrees from the university, and much more. This type of information was essential for building a trusted relationship between IBM and the university's leadership, with the university seen as a complex enterprise itself. Also, it was important for the IBM PEP to know if the university was an IBM SPSS analytics customer or an IBM Maximo asset management customer, or even an IBM supercomputer customer. To build a culture of trust around long-term UI relationships, as well as to jointly seek out opportunities to maximize win-win impact in areas of mutual interest, both quantitative and qualitative, short-term and long-term UI relationship information was essential.

Furthermore, the IBM GUP team of hundreds of dedicated professionals around the world, sometimes embedded in a research center at the university itself, ensured that small resource & small time-scale programs were effectively used with faculty and students in key technology and business areas. For example, local GUP teams could help faculty and students get access to IBM Watson in the cloud, allowing engineering school teams to prototype cognitive solutions, business school teams to run cognitive solution business plan competitions, and social science teams to explore the human-side of adopting advanced cognitive solutions.

The following highlight box summarizes years of experience from IBM GUP programs into a few concise bullets that may be helpful to other industry players looking to transform or expand their UI practices.

Highlight: Metrics, Incentives, Rewards and Culture

Opportunity

- Every change in business conditions and focus requires adjusting 6 R investment options and recalibrating metrics, incentives, rewards, and culture to fit the times.
- As the marginal cost of computing continues to drop, the full impact of AI in university-industry (UI) relationships will become more apparent.

Approach

- The key to culture is trust based on history of UI relationship as well as the collaborative practice of clearly defining problems of mutual strong interest that can only be solved working together to achieve win-win outcomes. Failures are mutual learning experiences.
- AI will not replace service providers (e.g., industry researchers, academic faculty, etc.), but trusted service providers who use AI effectively and ethically will replace those who don't.

Execution

- Based on business conditions and business focus, investments are carefully made in a mixture of small, medium, and large UI program options.
- Metrics are a mixture of quantitative and qualitative associated with 6 R program options that range from short-term, medium-term and long-term timescales.

Outcomes

- Success breeds success. Metrics associated with increasing flows of talent, publicity, publications and commercial offerings that highlight the new knowledge co-created are the outcomes that matter most in re-inventing a culture of UI collaboration generation after generation.

5.5 Case Studies

This section will contrast various recent collaboration models by comparing motivations and metrics and examining how these models drive research and workforce training (readiness). The section will also focus on the innovative aspects of engagement models.

Collaborative models can be broadly categorized into small, medium, and large based on the number of faculty/students associated with the program. Coincidentally and not surprisingly, the three categories also map into the amount of funding attached to the projects/program. The classification is somewhat arbitrary in that boundaries are elastic but years of experience driving these programs suggest that it is reasonable to generalize these engagement models. Broadly speaking, we can also define small relationships as those that have investments of the order \$0–\$250K

per annum; medium range from \$250K to \$1M and big relationships go from about a million to several million dollars. The rare very large mega relationship could exceed \$100M. Let's look at these classes of UI relationships and examine their motivations and metrics.

5.5.1 *Small Programs*

Small sub \$250K relationships are almost always between a company researcher or two as well as a university professor, including his/her graduate student. They are primarily exploratory in nature and often tied to a thesis topic of mutual relevance. The duration is often 2–3 years to benefit a graduate student's tenure in the department. The metric is primarily conference papers and associated publications but more recently there is an emphasis on IP creation.

It is worth noting that when IP language creeps into the agreement, relationships tend to be more difficult and painful to establish. While clear to the researchers, both the university and industry legal teams often do not realize that exploratory programs are mostly precompetitive in nature and the IP created has limited value. As for metrics, noteworthy exceptions exist, such as when funding is used as a seed to generate proposals for government funding and the metric then is getting a proposal funded.

Another type of small relationship is where the funds are set aside for internships, both summer and longer terms. These are primarily to identify students for potential future hiring. Here, the metric is hiring successes and the development of an ongoing pipeline for talent. At IBM we have successfully used long-term (9–12 months) graduate student internships from European universities that encourage semester long internships and often make industrial experience part of the master's degree requirements. Typically, these arrangements are very difficult to put in place with larger US research universities since there is no provision for co-op like engagements for students in graduate school, only fully paid graduate fellowships, which are significantly more costly to the industry partner. Here are some examples of small-scale programs that have worked well in the recent past.

- A first example is IBM's relationship with the **University of Turin** in Italy, offering semester and year-long internship opportunities to students in their graduate Master's program. The employment opportunity made the program coveted by students. As a result, the university could do a great job of identifying the best students through a competitive application process, becoming a win-win model. IBM benefited from excellent students, who spent 9–12 months in an internship-like program, often resulting in publications and/or IP. European universities are especially attractive for such programs because many require industrial experience as part of the graduate degree. In addition, countries like Italy even fund internships.

- A second example is a project on organocatalysis between a researcher at IBM and a professor at **Stanford**, aiming to jumpstart a field of catalysis that does not need heavy metals like platinum. This was a successful project, yielding a family of organic molecules that were effective in catalyzing polymerization and depolymerization reactions that led to discovering a way to depolymerize PET, offering a way to recycle plastic bottles and polyester fabric used increasingly in performance clothing.
- A third example is the **Notre Dame–IBM Technology Ethics Lab** (Moran Walton, 2024). Researchers will work together to study the ethical challenges emerging at the research frontier of large language models (LLMs).

5.5.2 *Medium Programs*

Medium programs often involve several professors and graduate students from a school within a university. Typically, such engagements are put in place to jumpstart an area of research in the industrial partner lab and where the department or school has demonstrated thought leadership in that field of study. Metrics, in this case, beyond the usual publications and conference papers are the time it takes to establish a center of competence in the industrial partner's lab and the graduate student pipeline to grow the program/business.

Rensselaer-IBM Artificial Intelligence Research Collaboration (AIRC) The Future of Computing Institute (FOCI) is advancing the science of artificial intelligence and enabling the use of AI and machine learning in research investigations, innovations, and applications of joint interest to both Rensselaer and IBM (Rensselaer, 2024). The success of such broad engagements depends on the commitment each partner brings to the table. The AIRC program is driven by an IBM executive and the Research VP from RPI. The technical agenda is overseen by an IBM Fellow together with the Chief Information Officer from RPI. In addition, program managers are responsible for driving the program and measuring the output. The collaborative research program yielded nearly 50 publications. An innovative aspect of this program is the internship guarantee by IBM for a set of RPI students, which involves placement in multiple global IBM labs.

5.5.3 *Large and Very Large Programs*

Large engagements (>\$1M) and very large engagements (>\$10M) share similar characteristics and will be discussed together in this section. Such large investments have ambitious goals that seek to either birth a new field of study or substantially and aggressively move a nascent field forward. In the case of semiconductor technology, an aim is to bring R&D back to the United States to address supply chain

instabilities due to geopolitical tensions. Such large investments often require multiple stakeholders. Let's look at a few recent relationships struck between IBM Research and academic partners and walk through their goals, metrics, and innovative elements.

IBM-Illinois Discovery Accelerator Institute (IDAI) is a new model of academic-industry partnership designed to increase access to technology education and skill development to spur breakthroughs in emerging areas of technology, referred to as research thrusts (UIUC, 2021). These include Hybrid Cloud & AI, Quantum Computing, Materials Discovery, and Sustainability, each aiming to accelerate the discovery of solutions to complex global challenges. At its inception in Fall 2021, the Institute's goal was to guide and support relevant projects within these thrusts. In 2023, the Institute entered a new phase by refocusing its activities on the development of an open high-performance computing platform to support AI-infused applications. This platform will link classical and quantum computing at the infrastructure layer while providing optimized AI model training and inferencing.

IDAI will enable the community to better cope with the generative AI (Gen AI) explosion while providing end-to-end security. It will also provide leverage for a wide range of strategic downstream activities within the Institute's thrusts, including AI for code, sustainability, and accelerated materials discovery. While the development of the platform remains an aspirational goal, it has steered the selection of projects that are funded by the Institute. A culture of innovation is enabled by open exchange of ideas and a high level of engagement and collaboration between faculty members, students, and IBM researchers. These are measurable through joint publications, number of graduate student theses that were shaped by work supported by the Institute, number of high-bandwidth collaborations enabled through mechanisms like externships, postdoctoral and sabbatical assignments at IBM. Another measure is IBM Researchers giving invited lectures, teaching classes, influencing the curriculum, and jointly pursuing external funding opportunities.

MIT-IBM Watson AI Lab is a community of scientists from MIT and IBM Research dedicated to pushing the frontiers of artificial intelligence and translating breakthroughs into real-world impact. Founded in 2017, the Lab works with industry to translate fundamental science into applications that solve immediate problems in the business world and beyond. The Lab currently manages a research portfolio of more than 80 projects, with an emphasis on data-driven, deep learning approaches to understanding language and the visual world as well as techniques for making large-scale AI systems more efficient and robust. The Lab is also developing AI systems for healthcare and a variety of decision-making applications.

In all of its work, the Lab is committed to building trustworthy and socially responsible AI systems. The center is located in one of the fastest-growing technology centers in the world: Kendall Square in Cambridge, Massachusetts. Across the street from MIT, down the road from Harvard, and situated in a dense cluster of the world's leading technology companies, Kendall Square is a vibrant ecosystem for innovators. In 2021, IBM Research team moved into the new offices on MIT's campus to be collocated with MIT faculty and students. The scale of the program requires several partners to come together to realize the lofty goals of working on

real-world challenges. The member companies have seats on the advisory board and influence the research portfolio with invaluable domain knowledge.

By grounding their R&D in the peer-reviewed science of the lab, participating companies ensure a strong foundation for innovation. Since the inception of this initiative, several industry partners have joined the program to leverage the knowledge of how one could apply AI techniques to their businesses. Large participating companies include: Shell, Evonik, Wells Fargo, Boston Scientific, Nexlore, and Woodside. The program's success is measured by metrics such as the number of members who have signed up and getting value. The membership benefits include very attractive IP Terms, IBM Support, Influence and Institutional Learning (MIT, 2024). To further expand on the agreement terms and to appreciate the value of the engagement model, here are some nuggets that can be gleaned from the **MIT-IBM Watson AI Lab website**. For IP, they point out that much of the open-source code in the marketplace can only be licensed for noncommercial use, but IP developed in this engagement can be used to build proprietary technology for business impact. In addition, the scientists from IBM Research will work with the member companies to develop strategy and pathways to scaling. They also point out that IBM can use their deep strengths in AI and their knowledge of businesses to integrate domain knowledge into their AI technologies. Finally, the engagement model also allows access to workshops, tutorials and webinars so the companies can develop internal AI skills for the future.

IBM Research—UAlbany's College of Nanoscale Science and Engineering is a cutting-edge facility located at the heart of New York's Tech Valley. It is owned and operated by NY Creates (NYCreates, 2024), one of the nation's largest and most advanced nonprofit semiconductor R&D facilities. The Albany NanoTech Complex is now a state-of-the-art campus that brings together industry leaders, academia and international partners to develop next-generation chips and chip fabrication processes. This facility has been a hub for major advancements and breakthroughs in chip technology for more than two decades.

With over 20 years of partnership between IBM, the State of New York, and many academic institutions across the country, this collaboration focuses on advanced semiconductor research and development. Leveraging state-of-the-art equipment and decades of expertise, IBM Research Albany continues to pioneer breakthroughs in materials science, nanotechnology, and chip design. Its interdisciplinary approach brings together scientists, engineers, and industry leaders to tackle some of the most pressing challenges in computing and technology. The site offers a fully-integrated research, development, prototyping, and educational facility that provides strategic support through outreach, technology acceleration, business incubation, pilot prototyping, and test-based integration support for onsite corporate partners including IBM, GlobalFoundries, Samsung, Applied Materials, Tokyo Electron, ASML and Lam Research, as well as other "next-generation" nanotechnology research activities, including hands-on internships for students along with career opportunities.

While Albany Nanotech has a distinct research focus, it has a large workforce development aspect that motivates significant investment from the state of New York.

The large collection of industrial partners helps targeted workforce training relevant to the semiconductor industry in the USA. This innovative business model brings together industry, academia, state government, and federal government and is a model worthy of emulation for national centers of excellence.

The parent entity NY CREATES aims to further accelerate the growth of high-value, high-technology companies, by attracting entrepreneurs to New York, supporting them through ideation, incubation, and investment phases. NY CREATES continues the long-standing push to provide an invigorating environment for high technology industries—which in the past 20-plus years has seen the establishment of IBM Research’s CMOS technology development facility, Tokyo Electron Ltd’s Technology Center America in Albany, Applied Materials META Center, GlobalFoundries’ flagship fab in Malta (NY), Danfoss’s power electronics packaging facility at Marcy (NY), and Wolfspeed’s announced expansion. Since inception, capital investment at the Albany NanoTech Campus has exceeded \$15 billion. The current operating budget for NY CREATES is approximately \$300 M/year and growing. The most recent innovation is nanosheets, which paves a path to the 2-nanometer (nm) node (Frougier & Guo, 2021).

5.5.4 Workforce Training Commentary

A couple of major changes in the technology landscape in the last few years has brought the issues of future workforce skill gaps and training to the forefront. The first driver of change is the increased urgency to translate technologies from the lab to the fab. Traditionally, exploratory research is the domain of academia, whereas industry had to grapple with the transfer of technology to the commercial domain. Many innovations at academic institutions never saw the light of day because there was no clear way to connect IP created in academia with industry needs. In rare instances when an industry-sponsored research yields actionable IP, there is a path forward.

There is a fundamental skill gap for traditional translational research. At academic institutions, there is little motivation to go beyond creating IP and publishing papers and at industrial research labs the skill to take research innovations to hardened products is in short supply. IBM has been very successful in creating IP as evidenced by holding a record for number of US Patents issued for nearly three decades from 1990 to 2020. That IP leadership has not been matched by an equally successful product innovation. Traditionally, IBM Research staffing has been weighted toward PhDs, with the ratio of PhDs to BS/MS being approximately 70/30. More recently, the research division has made a concerted effort to hire more MS and BS graduates to bring that ratio closer to 50/50, hoping to bolster the success of translation research. The main idea behind this workforce make-up shift is to have more engineers to help take ideas to early prototypes.

A second landscape shift is the increase in innovation at the intersection of disciplines. This has presented challenges to academic institutions, where departments

tend to be very siloed and degree requirements are biased towards coursework in a single department. Many leading universities have created centers of excellence that draw on faculty and students from across multiple departments. These are starting to bear fruit. With recent major advances in AI, there is a pressing need to integrate researchers from computer sciences and physical sciences and increasingly from social sciences. Such integrated programs will likely be created at industrial labs but the workforce training burden will still fall on academic institutions. This will have to be addressed not just by four-year degree programs but also by community colleges and high schools. The need for individuals with so-called “T-shaped skills, broad communication skills across disciplines and deep problem solving skills in a home discipline” continues to be both a priority and a challenge in academia and industry (Donofrio & DeMarco, 2022).

The burden of multidisciplinary training also falls on the federal government through funding agencies like the National Science Foundation (NSF), which is making admirable progress. It is worth calling out the Technology, Innovation and Partnership (TIP) directorate at the NSF (NSF, 2024a). TIP is a crosscutting platform that collaboratively integrates NSF’s existing directorates and fosters partnerships—with government, industry, academia, and nonprofits.

Keeping translational research effectiveness at the core, NSF’s Convergence Accelerator program is a commendable innovative program where industry, for the first time, can take on the role of a prime grantee in an NSF proposal (NSF, 2024b). Another program from NSF is the Engineering Research Visioning Alliance (ERVA), created to jump start a conversation on the direction of engineering education in the US (ERVA, 2024). The mission of ERVA is to identify and develop bold and transformative new engineering research directions and to catalyze the engineering community’s pursuit of innovative, high-impact research that benefits society.

5.6 Takeaways and Future Strategies

Due to IBM’s global scale across all industries working with thousands of universities in dozens of nations, there are many lessons to be learned from its metrics, incentives, rewards, and culture for impact. The key to successful impact has always been a foundation of trust and mutual respect—along with a strong UI collaboration culture with a long history of industry transforming success stories. Another key ingredient has been learning from experience to reassess and refresh UI programs to address changing market needs, such as customers desire for more open source-based innovation (Gil 2023).

Win-win UI programs can be co-created, with distinct but aligned metrics, incentives and rewards, and successfully executed together across small, medium, and large investments, measured both in duration and resources. To summarize, we recommend these metrics, each aligned with company strategy and the scale of investment:

- Research: Number of joint publications, scholarly and press releases, and collaborative research agreements started, ongoing, and ending.
- Readiness: Number of students and faculty taking IBM training and certification courses.
- Recruiting: Tracking both annual interns and full-time hires by university and degree, as well as tracking long-term trends.
- Revenue: Hardware, software, and service sales to universities.
- Responsibility: Number of employees in adjunct faculty, advisor board, industry mentor roles.
- Regions: Number of public-private partnerships.
- Refresh: Number of new programs started, ranking of success of existing programs, and old programs discontinued to adjust to changing market needs.

Future strategies will certainly be shaped and repeatedly reshaped by advancing AI capabilities. For example, as the marginal cost of computing drops lower, future UI strategies have both challenges to address and opportunities to exploit more automation of processes and augmentation of people. As succinctly summarized in Rouse and Spohrer (2018), one aspect of AI adoption seems clear:

AI will not replace service providers (e.g., industry researchers, academic faculty, etc.), but trusted service providers who use AI effectively and ethically will replace those who don't.

To be competitive, people and enterprises will need to upskill with AI. Not only that, UI interactions will profoundly change as industry researchers and academic faculty develop AI digital twins of themselves (Spohrer, 2024a, 2024b). Much work remains to be done to understand the coming impact of AI on UI relationships.

References

- Ankrah, S., & Omar, A. T. (2015). Universities–industry collaboration: A systematic review. *Scandinavian Journal of Management*, 31(3), 387–408.
- Aspray, W., & Williams, B. O. (1994). Arming American scientists: NSF and the provision of scientific computing facilities for universities, 1950–1973. *IEEE Annals of the History of Computing*, 16(4), 60–74.
- Clauss, T., Kesting, T., & Franco, M. (2024). Innovation generation through formalisation and fairness in university–industry collaboration. *Technovation*, 134, 103049.
- Cohen, M., Fernandes, G., & Godinho, P. (2024). Measuring the impacts of university–industry R&D collaborations: A systematic literature review. *The Journal of Technology Transfer*, 1–30.
- Compagnucci, L., & Spigarelli, F. (2020). The third mission of the university: A systematic literature review on potentials and constraints. *Technological Forecasting and Social Change*, 161, 120284.
- D'Este, P., & Perkmann, M. (2011). Why do academics engage with industry? The entrepreneurial university and individual motivations. *The Journal of Technology Transfer*, 36, 316–339.
- Donofrio, N., & DeMarco, M. (2022). *If nothing changes, nothing changes: The Nick Donofrio Story*. Houndstooth Press.
- ERVA. (2024). *NSF engineering research visioning alliance*. Website Home. <https://www.erva-community.org>

- Etzkowitz, H. (2016). The entrepreneurial university: Vision and metrics. *Industry and Higher Education*, 30(2), 83–97.
- Frougier, J., & Guo, D. (2021). *Introducing the world's first 2 nm node chip: IBM research's second-generation nanosheet technology has paved a path to the 2 nm node, produced on a 300 mm wafer*. IBM Research News. <https://research.ibm.com/blog/2-nm-chip>
- Gil, D. (2023, January 6). *Why IBM is no longer interested in breaking patent records—and how it plans to measure innovation in the age of open source and quantum computing*. Fortune. Commentary – Innovation. <https://fortune.com/2023/01/06/ibm-patent-record-how-to-measure-innovation-open-source-quantum-computing-tech/>
- Guenduez, A. A., Frischknecht, R., Frowein, S. C., & Schedler, K. (2024). Government-university collaboration on smart city and smart government projects: What are the success factors? *Cities*, 144, 104648.
- Hemmert, M., Bstieler, L., & Okamuro, H. (2014). Bridging the cultural divide: Trust formation in university–industry research collaborations in the US, Japan, and South Korea. *Technovation*, 34(10), 605–616.
- Kelly, J. E., & Hamm, S. (2013). *Smart machines: IBM's Watson and the era of cognitive computing*. Columbia University Press.
- Klofsten, M., Fayolle, A., Guerrero, M., Mian, S., Urbano, D., & Wright, M. (2019). The entrepreneurial university as driver for economic growth and social change - Key strategic challenges. *Technological Forecasting and Social Change*, 141, 149–158.
- Krishna, A. (2017). *The journey of Enterprise software to hybrid cloud*. IBM Investor Briefing. https://www.ibm.com/investor/att/pdf/2017_Investor_Briefing_The_Journey_of_Enterprise_Software_to_Hybrid_Cloud_charts.pdf
- Lella, G., Fischetto, A., Cesarotti, V., Spohrer, J. C., Ren, G., & Leung, Y. T. (2012). Universities as complex service systems: External and internal perspectives. In *Proceedings of 2012 IEEE International Conference on Service Operations and Logistics, and Informatics* (pp. 422–427). IEEE.
- Link, A. N., & Siegel, D. S. (2005). Generating science-based growth: An econometric analysis of the impact of organizational incentives on university–industry technology transfer. *European Journal of Finance*, 11(3), 169–181.
- MIT. (2024). *Membership: Finance. Construction. Healthcare. Energy. Consumer technology. International development. We're partnering with global leaders harnessing AI to change the world*. MIT IBM Watson AI Lab Membership Webpage. <https://mitibmwatsonailab.mit.edu/about/membership/>
- Moran Walton, L. (2024). *Notre Dame faculty and IBM research partner to advance research in ethics and large language models*. Tech Ethics Lab News. <https://techethicslab.nd.edu/news/notre-dame-faculty-and-ibm-research-partner-to-advance-research-in-ethics-and-large-language-models/>
- NSF. (2024a). *Technology, innovation and partnerships: A directorate at the U.S. National Science Foundation*. Website Home. <https://new.nsf.gov/tip/latest>
- NSF. (2024b). *Convergence accelerator*. Website Home. <https://new.nsf.gov/funding/initiatives/convergence-accelerator>
- NYCreates. (2024). *NYCreates: Center for Research, economic advancement, technology, engineering and science*. Website Homepage. <https://ny-creates.org>.
- O'Dwyer, M., Filieri, R., & O'Malley, L. (2023). Establishing successful university–industry collaborations: Barriers and enablers deconstructed. *The Journal of Technology Transfer*, 48(3), 900–931.
- Palmisano, S. J. (2006). The globally integrated enterprise. *Foreign Affairs*, 85, 127–136.
- Park, H. W., & Leydesdorff, L. (2010). Longitudinal trends in networks of university–industry–government relations in South Korea: The role of programmatic incentives. *Research Policy*, 39(5), 640–649.
- Perkmann, M., Neely, A., & Walsh, K. (2011). How should firms evaluate success in university–industry alliances? A performance measurement system. *R&D Management*, 41(2), 202–216.

- Piva, E., & Rossi-Lamastra, C. (2013). Systems of indicators to evaluate the performance of university-industry alliances: A review of the literature and directions for future research. *Measuring Business Excellence*, 17(3), 40–54.
- Rajalo, S., & Vadi, M. (2017). University-industry innovation collaboration: Reconceptualization. *Technovation*, 62, 42–54.
- Rensselaer. (2024). *Future of computing research collaboration*. Home. <https://airc.rpi.edu>
- Rouse, W. B., & Spohrer, J. C. (2018). Automating versus augmenting intelligence. *Journal of Enterprise Transformation*, 8(1–2), 1–21.
- Rybnicek, R., & Königsgruber, R. (2019). What makes industry–university collaboration succeed? A systematic review of the literature. *Journal of Business Economics*, 89(2), 221–250.
- Sassi, M., & Mshenga, P. M. (2024). Unlocking the potential of university-industry collaborations in African higher education: A comprehensive examination of agricultural faculties. *Industry and Higher Education*. <https://doi.org/10.1177/09504222241254694>
- Spohrer, J. C. (2010). IBM'S university programs. *IEEE Computer*, 43(8), 102–104.
- Spohrer, J. C. (2013, May 14). *What's up at IBM? University Programs! The 6 R's helping to build a smarter planet: Research, readiness, recruiting, revenue, responsibility, regions*. Slideshare. <https://www.slideshare.net/slideshow/ibm-up-external-20130514-v11/21175603>
- Spohrer, J. (2017). IBM'S service journey: A summary sketch. *Industrial Marketing Management*, 60, 167–172.
- Spohrer, J. (2024a). AI upskilling and digital twins: A service science perspective on the industry 4.0 to industry 5.0 shift. In *Industry 4.0 to industry 5.0: Explorations in the transition from a techno-economic to a socio-technical future* (pp. 79–92). Springer Nature.
- Spohrer, J. (2024b). *Personal AI digital twins: The future of human interaction?* EIT Digital. <https://www.eitdigital.eu/newsroom/grow-digital-insights/personal-ai-digital-twins-the-future-of-human-interaction/>
- Tijssen, R. J., Yegros-Yegros, A., & Winnink, J. J. (2016). University–industry R&D linkage metrics: Validity and applicability in world university rankings. *Scientometrics*, 109, 677–696.
- UIUC. (2021). *IBM-Illinois discovery accelerator institute (IIDA)*. University of Illinois Urbana-Champaign (UIUC) - The Grainger College of Engineering. Research Institute Website Homepage. <https://discoveryacceleratorinstitute.grainger.illinois.edu>
- Wurth, B., MacKenzie, N. G., & Howick, S. (2024). Not seeing the forest for the trees? A systems approach to the entrepreneurial university. *Small Business Economics*, 63, 1–24.

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